

Townsend 3.6-3.8

Questions for Gallicchio

How do we calculate the probability of higher spin things happening without matrix algebra? Is there nice bra ket stuff?

3.6 The Spin $\frac{1}{2}$ Eigenvalue Problem

Spin angular momentum: $\hat{S}_{x,y, \text{ or } z}$

Regular Angular Momentum: $\hat{L}_{x,y, \text{ or } z}$

Pauli spin matrices $\hat{S} \rightarrow \frac{\hbar}{2}\sigma$, where S is the vector sum of $\hat{S}_{x,y,z}$ and σ is the same for $\sigma_{x,y,z}$. Eigenvalues of $n\hbar$.

Spin 1 particles

3 eigenvalues - $\hbar, 0, -\hbar$

kets are given by $|1, 1\rangle$ or $|1, 0\rangle$ or $|1, -1\rangle$

We just use 3d matrix algebra